Task 1: Problem Framing & LitMap

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1. Methodology Card

Topic

Robustness of text-based phishing e-mail detectors to phishing text paraphrased by large language models (LLMs).

Motivation: a summary of the FBI IC3 report puts 2025 cybercrime losses at $20.9 billion, 26% above the year before [13]. Studies suggest LLM-written phishing can match human-written phishing in click-through [4], [5].

Object and subject

Object (broad system)	Subject (specific metric / algorithm)
Automatic text-based phishing e-mail detectors: classical machine learning, fine-tuned transformers and LLMs.	The change in recall (and F1) at a fixed false-positive rate when original phishing e-mails are replaced by their LLM-paraphrased versions.

Research questions (FINER), hypotheses and variables

RQ	Question	H0	H1
RQ1 (comparative)	How much do recall and F1 of three detectors (TF-IDF + logistic regression, DistilBERT, zero-shot LLM) differ between original and LLM-paraphrased phishing e-mails?	Recall does not change: $\Delta\text{recall} = 0$ for each detector.	Recall drops for each detector: $\Delta\text{recall} > 0$.
RQ2 (causal)	Does LLM paraphrasing reduce the probability of detection at FPR = 1%, after controlling for dataset, e-mail length and detector type?	Text version has no effect: $\beta(\text{paraphrased}) = 0$.	Paraphrasing lowers detection odds: $\beta(\text{paraphrased}) < 0$.
RQ3 (optimisation)	Does adding paraphrased phishing e-mails to the training set restore recall to within 2 percentage points (p.p.) of the baseline, without raising FPR by more than 1 p.p.?	Recall stays more than 2 p.p. below baseline, or FPR rises by more than 1 p.p.	Recall ≥ baseline − 2 p.p. and $\Delta\text{FPR} \le 1$ p.p.

RQ	Independent variable	Dependent variable	Controlled variables
RQ1	Text version (original / paraphrased); detector type	Recall at FPR = 1%; F1; ROC-AUC	Dataset, train/validation/test split, threshold rule, random seeds
RQ2	Text version (original / paraphrased)	Detection outcome (0/1) at FPR = 1%	Detector type, dataset (source), log e-mail length; errors clustered by e-mail
RQ3	Training regime (plain / augmented with paraphrases)	Recall on paraphrased test e-mails; FPR	Detector type, dataset, split, seeds, paraphrasing model and prompt

FINER check. Feasible: the corpora are public and the models run on a free GPU. Interesting: AI-written phishing performs about as well as expert-written phishing [5]. Novel: no study found measures the paraphrasing effect at a fixed false-positive rate with controls (Section 3). Ethical: only existing public corpora are rewritten, and nothing is sent to people. Relevant: the results bear on how e-mail filters are built and tested.

Aim and tasks

Aim: to measure how well text-based phishing detectors withstand LLM paraphrasing, and to test whether training on paraphrases removes the loss.

1. Collect, clean and de-duplicate open e-mail corpora, then fix the train/validation/test split.

2. Paraphrase the phishing e-mails with an open-weights LLM and filter out poor rewrites.

3. Train the three detectors on original data and set the decision threshold at FPR = 1% on validation data. This is the baseline.

4. Measure the drop in recall and F1 on paraphrased test e-mails and test H0 and H1 for RQ1 and RQ2.

5. Retrain with paraphrase augmentation and test H0 and H1 for RQ3.

2. Related Work Matrix

Source	Method	Data / sample	Metrics	Key result	Limitation
[2] Lain et al., 2024	Field experiment + interviews	4,554 employees; 25 interviews	Clicks; dangerous actions	Most of the gain comes from the reminder; delayed training works as well as immediate	One organisation; no detectors
[3] Ho et al., 2025	Randomised experiment, 8 months	19,789 employees	Failure rate; odds ratio	Embedded training OR = 0.905; annual training shows no association	One organisation (healthcare)
[5] Heiding et al., 2024	Human-subject experiment; LLM as attacker	101 participants	Click-through rate	AI 54% = expert 54%; control 12%	Small sample; detectors not evaluated
[7] Bukhsh et al., 2025	Survey with simulated e-mails; logistic regression	164 respondents; 43 e-mails	Odds ratios; mean susceptibility	Urgency has the largest effect; women OR = 0.391	Convenience sample; self-report; one country
[8] Koide et al., 2024	GPT-4-based detector	Dataset not named in abstract	Accuracy	99.70%	Dataset and other metrics not reported
[9] Thapa et al., 2025	Quantized LLMs vs ML/DL; zero/few-shot	"Curated" dataset, not described	Accuracy; VRAM	LLMs trail ML/DL; LLM-rephrased e-mails degrade detectors	No numeric loss reported; data not described
[10] Ganiuly & Smaiyl, 2025	DeBERTa-v3, LLaMA-3-8B, GPT-4	Five corpora (incl. Enron-Spam, Nazario, CSDMC2010)	TCI; CDS; accuracy	Trust ranking GPT-4 > LLaMA > DeBERTa differs from accuracy ranking	No paraphrased e-mails; no figures in abstract
[11] Kuikel et al., 2025	LLMs; explanation consistency	Not stated	CC SHAP; accuracy	Llama more consistent but less accurate	Dataset not stated; no attack robustness
[12] Uddin et al., 2024	DistilBERT + LIME / Transformer Interpret	Dataset not named	Not named	High accuracy and explainability	Metrics and dataset not named

The matrix draws on abstracts and first pages. Check each entry against the full paper before submitting.

3. Literature Synthesis and Research Gap

Recent phishing research follows two lines. One asks why people fall for phishing and whether training helps. The other asks what generative AI changes for attackers and defenders. In a randomised trial with 19,789 employees of a healthcare organisation, embedded training cut the odds of failing a simulated phish by only about 9.5% (OR = 0.905), and annual awareness training showed no association with failure [3]. At a Swiss company, most of the benefit came from the reminder that a test had happened, not from the material, which employees rarely read [2]. An earlier study of 14,733 employees found that embedded training did not improve resilience and may have lowered it [1]. On the attacker side, LLMs now write phishing that works: fully automated AI e-mails and expert-written ones both reached a 54% click-through rate among 101 participants, against 12% for a generic control [5].

Work on detection is harder to compare. Uddin et al. fine-tune DistilBERT and explain its predictions with LIME, and report high accuracy without naming their metrics [12]. ChatSpamDetector, a GPT-4 system, reaches 99.70% accuracy on a dataset its abstract does not describe [8]. Quantized LLMs trail classical machine-learning and deep-learning models in accuracy [9], and how far a model can be trusted, judged by calibration, does not follow its accuracy [10], [11]. Ganiuly and Smaiyl test several corpora (including Enron-Spam and Nazario) under one protocol, but they do not use paraphrased e-mails [10]. Human-subject studies point to cues such as urgency [7] and to attention and security habits [6], though with small samples (N = 164 and N = 187).

Two gaps remain. First, the effect of LLM paraphrasing on detectors is barely quantified. Thapa et al. report that LLM-rephrased e-mails degrade both ML and LLM detectors, but the abstract gives no figures [9]. No study measures the drop at a fixed false-positive rate while controlling for dataset, e-mail length and model type. Second, several papers omit the dataset or report only accuracy, so their numbers cannot be compared.

This study addresses the first gap. On open corpora, it measures how much recall falls at a fixed false-positive rate when original phishing e-mails are replaced by LLM paraphrases, for a TF-IDF and logistic regression model, a fine-tuned DistilBERT and a zero-shot LLM. It then tests whether training on paraphrased e-mails brings recall back. The result is a reproducible baseline for text-based detectors facing AI-assisted phishing.

4. LitMap and Zotero

Zotero collection (Task1_Moldassanov): [insert shared link]

LitMap (Connected Papers, ResearchRabbit or Litmaps): [insert screenshot and public link]

References

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[4] F. Heiding, B. Schneier, A. Vishwanath, J. Bernstein, and P. S. Park, "Devising and detecting phishing: Large language models vs. smaller human models," arXiv:2308.12287, 2023.

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[10] D. Ganiuly and A. Smaiyl, "Trustworthiness calibration framework for phishing email detection using large language models," arXiv:2511.04728, 2025.

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[12] M. A. Uddin, M. Mahiuddin, and I. H. Sarker, "An explainable transformer-based model for phishing email detection: A large language model approach," arXiv:2402.13871, 2024.

[13] B. Thies, "FBI IC3 report: Losses hit $20.9 billion due to ATO, phishing, fraud," SpyCloud, Apr. 2026. [Online]. Available: https://spycloud.com/blog/fbi-internet-crime-report-2025/